



Expense Manager

A modern, secure, and responsive expense management web application built with **Next.js 14 (App Router)**, **Convex** for backend and database, and **Clerk** for authentication. This app empowers users to track, categorize, and manage their daily transactions with an intuitive and visually appealing interface.



Features

- 🔒 **Secure Authentication:** Seamless and secure sign-in/sign-up with Clerk.
- 📁 **Category Management:** Create, edit, and manage expense categories effortlessly.
- 📊 **Transaction Tracking:** Add, edit, delete, and view transactions with ease.
- 📅 **Yearly Expense History:** Visualize and analyze expenses year-by-year.
- ⚡ **Serverless Backend:** Powered by Convex for real-time database and serverless functions.
- 🎨 **Modern UI:** Built with Tailwind CSS, shadcn/ui, and Framer Motion for smooth animations.
- 🌙 **Dark Mode Support:** Toggle between light and dark themes for a comfortable user experience.
- 📱 **Responsive Design:** Optimized for both desktop and mobile devices.



Tech Stack

- [Next.js 14](#): React framework with App Router for optimized routing and API routes.
- [Convex](#): Serverless backend, database, and real-time API functions.
- [Clerk](#): Authentication and user management with a secure and customizable interface.
- [Tailwind CSS](#): Utility-first CSS framework for rapid and responsive styling.
- [shadcn/ui](#): Accessible and customizable UI components.
- [Framer Motion](#): Smooth and engaging animations for a polished user experience.



Project Structure

```
src/
├── app/
│   ├── api/
│   │   ├── transactions/route.ts    # API for transaction CRUD operations
│   │   ├── categories/route.ts      # API for category management
│   │   └── yearhistory/route.ts      # API for yearly expense history
│   ├── dashboard/                   # Main dashboard pages
│   └── auth/page.tsx                # Authentication page
├── components/                      # Reusable UI components
├── lib/
│   └── convexClient.ts              # Convex client configuration
└── convex/                          # Convex backend functions and schema
```



Setup Instructions

Prerequisites

- Node.js:** Version 18 or higher
- npm:** Version 8 or higher
- Convex CLI:** For managing the Convex backend

- **Clerk Account:** For authentication setup
- **Vercel Account:** For deployment (optional)

1. Clone the Repository

```
git clone https://github.com/your-username/expense-manager.git
cd expense-manager
```

2. Install Dependencies

```
npm install
```

3. Set Up Convex

Install the Convex CLI globally:

```
npm install -g convex
```

Initialize Convex in your project:

```
npx convex dev
```

Deploy Convex functions after making changes:

```
npx convex deploy
```

4. Set Up Clerk

1. Create a project in the [Clerk Dashboard](#).
2. Obtain your **Frontend API Key** and **Secret Key**.
3. Create a `.env.local` file in the project root and add the following:

```
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=your_publishable_key
CLERK_SECRET_KEY=your_secret_key
NEXT_PUBLIC_CONVEX_URL=your_convex_url
```

5. Run the Development Server

```
npm run dev
```

The app will be available at `http://localhost:3000`.

6. Deploy to Vercel



Deploy

Deploy your app effortlessly with Vercel:

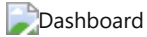
Screenshots & Demo

Authentication



Secure login and sign-up powered by Clerk.

Dashboard



Track and manage categories, transactions, and yearly history in one place.

Contributing

Contributions are welcome! Follow these steps to contribute:

1. **Fork the Repository:** Click the "Fork" button on GitHub.
2. **Create a Feature Branch:** `git checkout -b feature/your-feature-name`
3. **Commit Changes:** `git commit -m "Add your feature description"`
4. **Push to Your Fork:** `git push origin feature/your-feature-name`
5. **Submit a Pull Request:** Open a pull request with a clear description of your changes.

Please ensure your code follows the project's coding standards and includes appropriate tests.

License

This project is licensed under the [MIT License](#).

Happy expense tracking! 